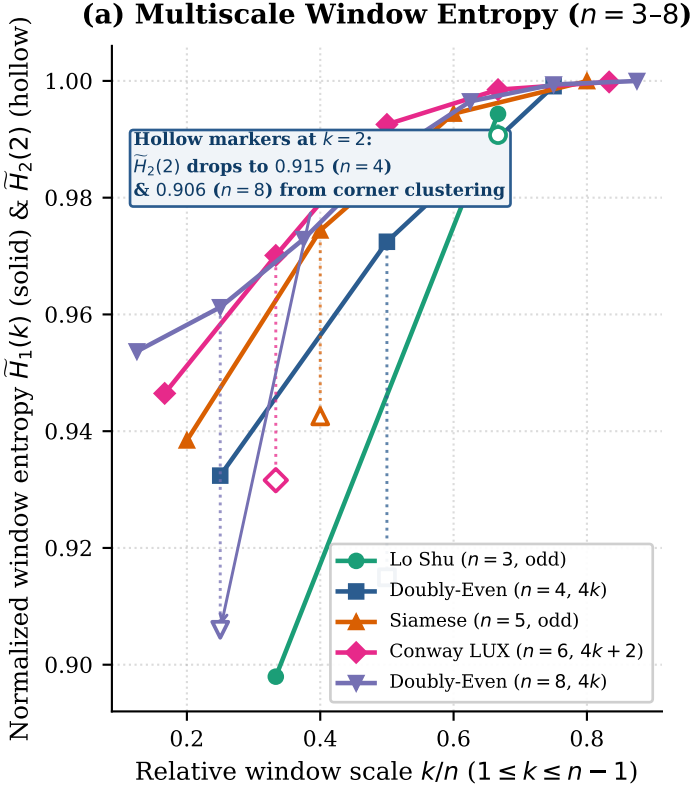
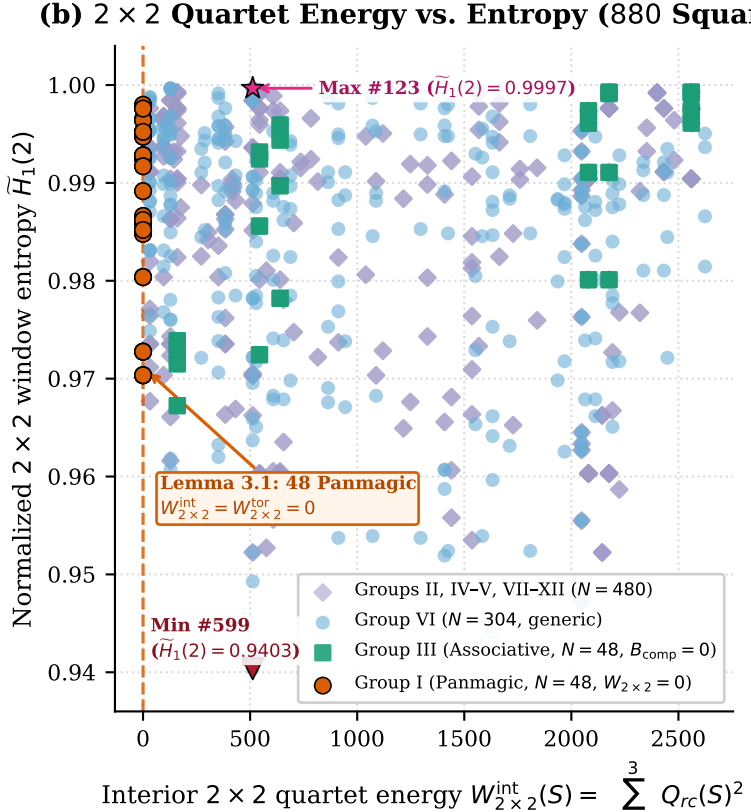
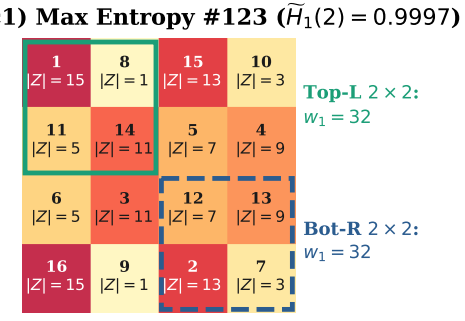
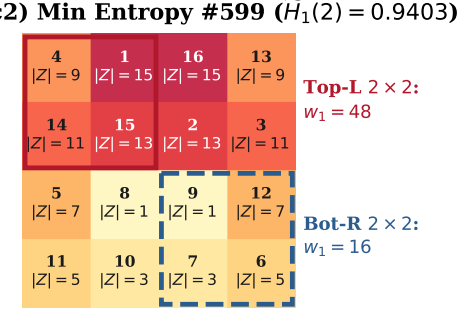

(a) Multiscale Window Entropy ($n = 3$–8)
Normalized window entropy $\widetilde{H}_1(k)$ (solid) & $\widetilde{H}_2(2)$ (hollow)
Relative window scale k/n $(1 \le k \le n-1)$
Hollow markers at $k = 2$:
$\widetilde{H}_2(2)$ drops to 0.915 ($n = 4$)
& 0.906 ($n = 8$) from corner clustering
Lo Shu ($n = 3$, odd)
Doubly-Even ($n = 4$, $4k$)
Siamese ($n = 5$, odd)
Conway LUX ($n = 6$, $4k + 2$)
Doubly-Even ($n = 8$, $4k$)


(b) 2×2 Quartet Energy vs. Entropy (880 Squares)
Normalized 2×2 window entropy $\widetilde{H}_1(2)$
Max #123 ($\widetilde{H}_1(2) = 0.9997$)
Lemma 3.1: 48 Panmagic
$W^{\text{int}}_{2\times2} = W^{\text{tor}}_{2\times2} = 0$
Min #599
($\widetilde{H}_1(2) = 0.9403$)
Groups II, IV–V, VII–XII ($N = 480$)
Group VI ($N = 304$, generic)
Group III (Associative, $N = 48$, $B_{\text{comp}} = 0$)
Group I (Panmagic, $N = 48$, $W_{2\times2} = 0$)
Interior 2×2 quartet energy $W^{\text{int}}_{2\times2}(S) = \sum_{r,c=1}^{3} Q_{rc}(S)^2$


(c1) Max Entropy #123 ($\widetilde{H}_1(2) = 0.9997$)
Top-L 2×2: $w_1 = 32$
Bot-R 2×2: $w_1 = 32$


(c2) Min Entropy #599 ($\widetilde{H}_1(2) = 0.9403$)
Top-L 2×2: $w_1 = 48$
Bot-R 2×2: $w_1 = 16$